

EXPERIMENT 28

Ransomware Behavior Assessment

AIM

To assess whether simulated file-system activity resembles ransomware behavior using multiple indicators.

ALGORITHM

1. Create a simulated list of file-system events.
2. Count rapid file modifications.
3. Count extension changes and renames.
4. Check whether several directories are affected.
5. Add risk points for each indicator.
6. Display the risk score and assessment.

CODE

```
events = [  
    ("09:00", "modify", "a.docx"),  
    ("09:01", "rename", "a.locked"),  
    ("09:01", "modify", "b.docx"),  
    ("09:02", "rename", "b.locked"),  
    ("09:03", "rename", "c.locked")  
]  
  
score = 0  
modifications = sum(1 for e in events if e[1] == "modify")  
renames = sum(1 for e in events if e[1] == "rename")  
  
if modifications >= 2:  
    score += 2  
if renames >= 2:  
    score += 2  
if any(".locked" in e[2] for e in events):  
    score += 2  
  
print("RANSOMWARE BEHAVIOR ASSESSMENT")  
print("Modified files:", modifications)  
print("Renamed files:", renames)  
print("Risk score:", score)  
  
if score >= 5:  
    print("Assessment: HIGH RISK - behavior resembles ransomware")  
else:  
    print("Assessment: LOW/MEDIUM RISK")
```

OUTPUT

```
Visual Studio Code

program_28.py
1  events = [
2      ("09:00", "modify", "a.docx"),
3      ("09:01", "rename", "a.locked"),
4      ("09:01", "modify", "b.docx"),
5      ("09:02", "rename", "b.locked"),
6      ("09:03", "rename", "c.locked")
7  ]
8
9  score = 0
10 modifications = sum(1 for e in events if e[1] == "modify")
11 renames = sum(1 for e in events if e[1] == "rename")
12
13 if modifications >= 2:
14     score += 2
15 if renames >= 2:
16     score += 2
17 if any(".locked" in e[2] for e in events):
18     score += 2
19
20 print("RANSOMWARE BEHAVIOR ASSESSMENT")
21 print("Modified files:", modifications)
22 print("Renamed files:", renames)
23 print("Risk score:", score)
24
25 if score >= 5:
26     print("Assessment: HIGH RISK - behavior resembles ransomware")
27 else:
28     print("Assessment: LOW RISK - behavior does not resemble ransomware")
29
```

```
TERMINAL
PS C:\Forensics> python program_28.py
RANSOMWARE BEHAVIOR ASSESSMENT
Modified files: 2
Renamed files: 3
Risk score: 6
Assessment: HIGH RISK - behavior resembles ransomware
```

RESULT

The program calculates a ransomware-behavior risk score from simulated file activity.